Statistical Business Analysis

Week 7 Project Report: Introduction to Statistics for Data Science

Executive Summary

This report applies statistics to two business datasets: a sales file with 100 orders (1 January to 9 April 2024) and a customer file with 500 customers. It calculates descriptive statistics, checks the shape of the data, measures correlations, runs 10 hypothesis tests, builds 95% confidence intervals and fits regression models.

- **Customers leave early and on short contracts.** 63.1% of customers with 12 months or less of tenure left, against 0.0% of those with more than 12 months. Month-to-month customers churn at 20.6% against 4.3% for one-year contracts (chi-square p = 9.6e-07). Both results stay significant after a strict multiple-test correction.
- **The average order is Rs.123,650** (95% confidence interval Rs.103,776 to Rs.143,525). Order values are skewed to the right (skewness 0.88) and not normally distributed, so the median (Rs.97,956) is a fairer "typical" value than the mean.
- **Order value is driven by quantity and price** (correlations 0.69 and 0.65). In this data Total_Sales is exactly Quantity x Price, so the regression confirms how the data was recorded rather than revealing a new relationship.
- **Some differences are only hints.** West orders look smaller (Rs.81,689 against Rs.138,394, p = 0.009) and customers who left paid more per month (p = 0.010), but neither stays significant after correcting for running 10 tests. They need more data before action.
- **No marketing-spend analysis was possible:** the data has no marketing-spend column, so the link between marketing spend and revenue could not be tested.

Top recommendations: (1) focus retention efforts on the first 12 months, (2) encourage month-to-month customers to move to longer contracts, (3) investigate small orders in the West region with more data, and (4) start recording marketing spend so its effect on revenue can be tested.

1. Data and Methodology

Sales data (sales_data.csv, saved in the project as business_data.csv): 100 orders with Date, Product, Quantity, Price, Customer_ID, Region and Total_Sales. **Customer data (customer_churn.csv):** 500 customers with Tenure, MonthlyCharges, TotalCharges, Contract, PaymentMethod, PaperlessBilling, SeniorCitizen and Churn (1 = left, 0 = stayed). Neither file has missing values or duplicate rows. The two files were analysed separately, because their customer IDs use different formats and cannot be matched reliably.

Methods (Python: pandas, numpy, scipy, statsmodels):

- Descriptive statistics: mean, median, mode, standard deviation, variance, range, quartiles, IQR, skewness and kurtosis.
- Distribution check: histogram, Q-Q plot and the Shapiro-Wilk normality test.
- Correlation: Pearson and Spearman correlation coefficients and the covariance matrix.
- Hypothesis tests at the 5% significance level (alpha = 0.05): Welch t-tests, one-way ANOVA (with Kruskal-Wallis and Mann-Whitney as checks) and chi-square tests of independence, with Cohen's d as the effect size.
- 95% confidence intervals: t-intervals for means, Welch interval for a difference, and Wilson intervals for proportions.

- Regression: ordinary least squares with R-squared, coefficient confidence intervals, and a residual check.
- Multiple-test control: a Bonferroni correction for the 10 tests.

A note on marketing spend

The project brief mentions the relationship between marketing spend and revenue. The datasets provided contain no marketing-spend column, so this relationship could not be analysed and no data was invented. Instead, the correlation and regression work uses the business metrics that exist: quantity, price and order value in the sales data, and tenure, monthly charges and churn in the customer data.

2. Descriptive Statistics

2.1 Sales data

Statistic	Total_Sales (Rs.)	Quantity	Price (Rs.)
Mean	123,650.48	4.78	25,808.51
Median	97,955.50	5.00	24,192.00
Mode	none (no repeats)	4.00	none (no repeats)
Std deviation	100,161.09	2.59	13,917.63
Minimum	6,540.00	1.00	1,308.00
Maximum	373,932.00	9.00	49,930.00
Q1 (25%)	39,517.50	2.75	14,965.25
Q3 (75%)	175,792.50	7.00	38,682.25
IQR	136,275.00	4.25	23,717.00
Skewness	0.88	0.07	-0.01
Kurtosis	-0.18	-1.20	-1.26

The average order is Rs.123,650, but half of all orders are below Rs.97,956. The standard deviation (Rs.100,161) is large, nearly as big as the mean, so order values vary a lot. No order value repeats, so there is no mode for Total_Sales. The most common quantity is 4 units, the most common product is Tablet and the most common region is North.

Order value by product

Product	Orders	Mean (Rs.)	Median (Rs.)	Std deviation (Rs.)
Laptop	24	162,050	113,720	116,570
Phone	20	142,970	123,922	107,192
Tablet	26	110,936	97,788	82,367
Headphones	15	92,269	61,624	96,177
Monitor	15	89,871	49,280	77,595

2.2 Customer data

Statistic	Tenure (months)	MonthlyCharges	TotalCharges
Mean	36.53	113.64	4,237.88
Median	37.00	115.00	4,182.50
Mode	3.00	115.00	4,023.00
Std deviation	20.67	51.80	2,260.62

Statistic	Tenure (months)	MonthlyCharges	TotalCharges
Minimum	1.00	20.00	159.00
Maximum	71.00	199.00	7,992.00
Q1 (25%)	19.00	67.00	2,237.25
Q3 (75%)	54.00	158.00	6,266.75
IQR	35.00	91.00	4,029.50
Skewness	-0.09	-0.09	-0.03
Kurtosis	-1.22	-1.21	-1.22

Customers stay 36.5 months on average (range 1 to 71 months). The three variables are close to evenly spread (skewness near 0). The churn rate is 10.6% (53 of 500 customers).

2.3 Distribution of order values

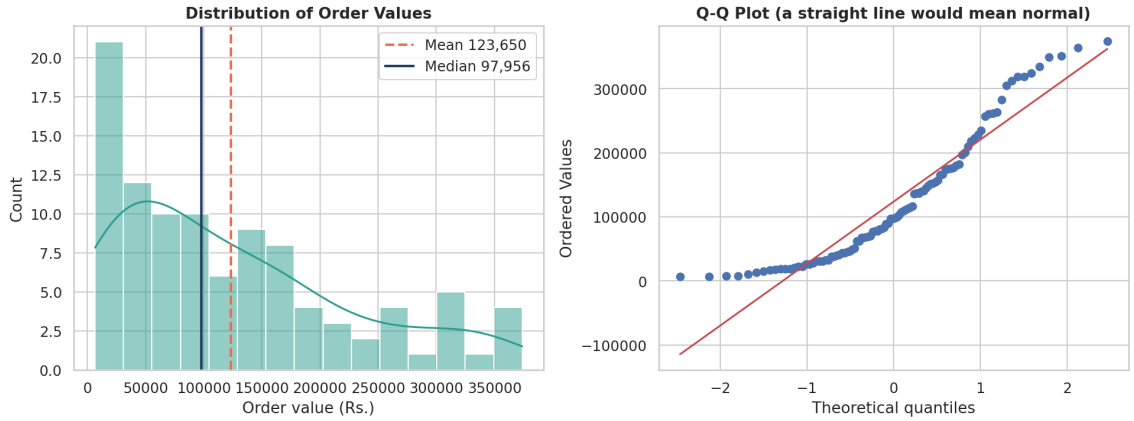


Figure 1: Histogram with mean and median (left) and Q-Q plot (right)

The Shapiro-Wilk test gives p = 1.3e-06, so the order values are not normally distributed. The histogram has a long tail to the right and the Q-Q points curve away from the straight line. Because of this, the report also uses non-parametric tests (Mann-Whitney and Kruskal-Wallis) as checks, and the results of the t-tests and ANOVA should be read with some care.

3. Correlation Analysis

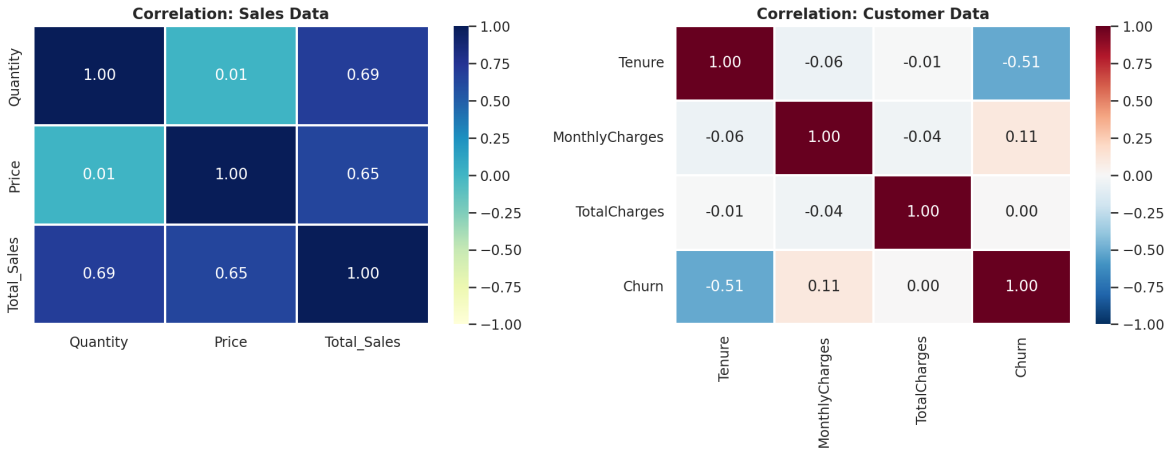


Figure 2: Pearson correlation heatmaps for the sales data (left) and the customer data (right)

Pair	Pearson r	p-value	Spearman rho
Quantity and Total_Sales	0.688	2.6e-15	0.691
Price and Total_Sales	0.646	3.9e-13	0.680
Quantity and Price	0.008	0.94	0.011
Tenure and Churn	-0.509	2.4e-34	-0.495
MonthlyCharges and Churn	0.107	0.016	0.106

- **Order value is strongly linked with quantity (r = 0.69) and price (r = 0.65).** Both links are highly significant. Quantity and price are unrelated to each other (r = 0.01), so each adds its own information.
- **Tenure and churn have a moderate negative link (r = -0.51):** the longer a customer stays, the less likely they are to leave.
- **MonthlyCharges and churn have a weak positive link (r = 0.11).**
- **Data quality warning:** in real billing data, TotalCharges should be close to Tenure x MonthlyCharges. Here the correlation between them is only -0.04, and TotalCharges is unrelated to tenure (r = -0.01). TotalCharges therefore looks unreliable and was not used in any test.
- Correlation does not prove cause and effect. The covariance matrix is in the notebook; correlation is easier to compare because it does not depend on the units.

4. Hypothesis Tests

Each test starts from a null hypothesis (H0, "no effect") and uses the p-value to decide. A p-value below 0.05 means the result would be unlikely if H0 were true. Because 10 tests were run, a Bonferroni-adjusted p-value (p x 10) is also shown: it is stricter and guards against results that look significant by luck.

#	Test and question	Result	p-value	Adjusted p (x10)	Decision
1	Welch t-test: West orders vs other regions	Rs.81,689 vs Rs.138,394; d = -0.58	0.0089	0.089	Significant (survives correction: No)
2	Welch t-test: Laptop orders vs other products	Rs.162,050 vs Rs.111,524	0.0609	0.609	Not significant
3	ANOVA: order value across 5 products	no clear difference	0.0934	0.934	Not significant
4	ANOVA: order value across 4 regions	no clear difference	0.0972	0.972	Not significant
5	Chi-square: churn vs contract type	churn 20.6% / 4.3% / 6.9%	9.6e-07	9.6e-06	Significant (survives correction: Yes)
6	Welch t-test: tenure, left vs stayed	6.0 vs 40.2 months; d = -1.92	9.0e-130	9.0e-129	Significant (survives correction: Yes)
7	Welch t-test: monthly charges, left vs stayed	129.8 vs 111.7; d = 0.35	0.0101	0.101	Significant (survives correction: No)
8	Chi-square: churn vs payment method	no link	0.144	1.00	Not significant
9	Chi-square: churn vs paperless billing	no link	0.829	1.00	Not significant
10	Chi-square: churn vs senior citizen	no link	0.795	1.00	Not significant

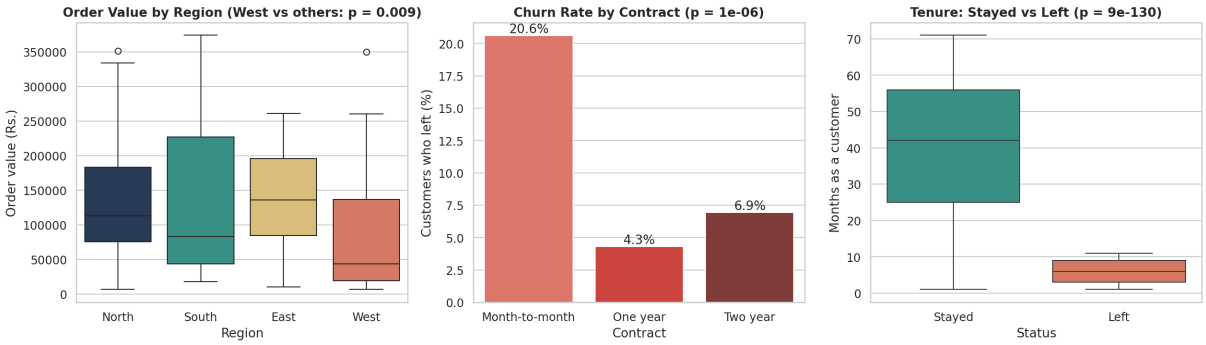


Figure 3: The main tests: order value by region, churn rate by contract, and tenure of customers who stayed or left

Interpretation

- **Strong evidence (Tests 5 and 6):** churn depends on the contract type, and customers who left had a far shorter tenure (6.0 against 40.2 months). These results are extremely unlikely to be chance and remain significant after correction.
- **Suggestive only (Tests 1 and 7):** West orders are about Rs.56,704 smaller on average (p = 0.009, medium effect size d = -0.58; the Mann-Whitney test agrees), and customers who left paid about 18.1 more per month. However, neither survives the correction for 10 tests. The West region was also picked after looking at the data, which makes its p-value optimistic.
- **No evidence of a difference (Tests 2, 3, 4, 8, 9, 10):** Laptop orders look larger (d = 0.51) but the data cannot rule out chance (p = 0.061), and the overall differences between products and between regions are not significant. With only 100 orders, small groups give little statistical power, so "not significant" does not prove that there is no difference.

5. Confidence Intervals

Quantity	Estimate	95% confidence interval	Margin of error
Average order value	Rs.123,650	Rs.103,776 to Rs.143,525	Rs.19,874
West minus other regions (average order)	Rs.-56,704	Rs.-98,529 to Rs.-14,880	Rs.41,824
Overall churn rate	10.6%	8.2% to 13.6%	2.7 points

Average order value by product

Product	Orders	Mean order	95% CI low	95% CI high
Laptop	24	Rs.162,050	Rs.112,827	Rs.211,274
Phone	20	Rs.142,970	Rs.92,803	Rs.193,137
Tablet	26	Rs.110,936	Rs.77,668	Rs.144,205
Headphones	15	Rs.92,269	Rs.39,008	Rs.145,530
Monitor	15	Rs.89,871	Rs.46,901	Rs.132,842

Churn rate by contract type

Contract	Customers	Left	Churn rate	95% CI
Month-to-month	170	35	20.6%	15.2% to 27.3%
One year	186	8	4.3%	2.2% to 8.3%
Two year	144	10	6.9%	3.8% to 12.3%

We can say with 95% confidence that the true average order value lies between Rs.103,776 and Rs.143,525; the margin of error is Rs.19,874 (16% of the mean), which is wide because orders vary a lot and there are only 100 of them. The product intervals overlap heavily, which agrees with the non-significant ANOVA. The West difference interval is entirely below zero, which supports Test 1 before the multiple-test correction. The month-to-month churn interval does not overlap with the one-year interval, so month-to-month churn really is higher.

6. Regression Analysis

Model 1: Total_Sales = intercept + b1 x Quantity + b2 x Price (ordinary least squares, n = 100).

Term	Coefficient	95% CI	p-value
const	-121,682.22	-140,871.10 to -102,493.33	4.1e-22
Quantity	26,430.84	23,773.56 to 29,088.12	9.6e-36
Price	4.61	4.12 to 5.10	1.3e-33

- **R-squared = 0.884** (adjusted 0.882): quantity and price together explain 88.4% of the differences in order value. The F-test is highly significant (F = 369, p = 4.4e-46).
- Each extra unit adds about Rs.26,431 to an order (95% CI Rs.23,774 to Rs.29,088), and each extra Rs.1 of price adds about Rs.4.61.
- **The model has a flaw.** Its smallest prediction is Rs.-64,268, which is impossible for an order value, and the residuals form a U-shaped curve (Figure 4, right). This happens because Total_Sales is exactly Quantity x Price in every row, and a straight line cannot copy a multiplication.

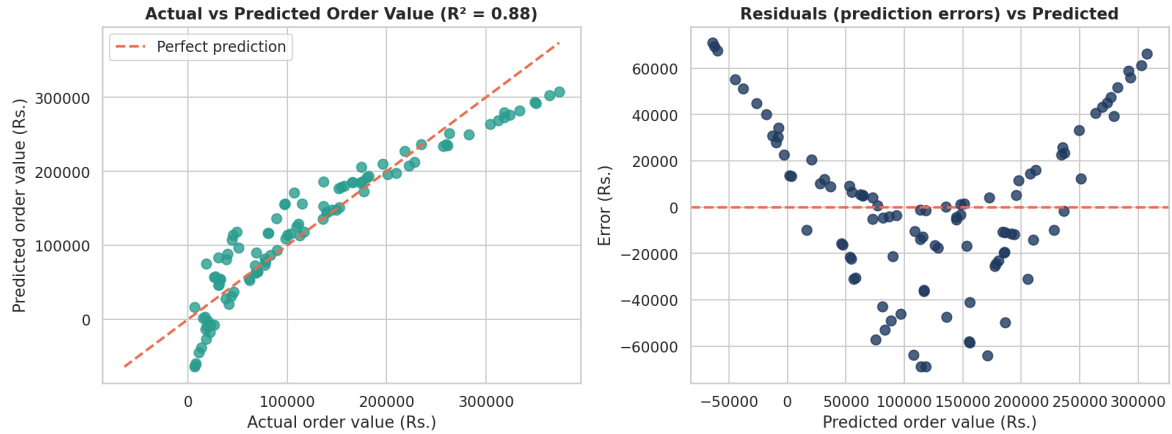


Figure 4: Actual against predicted order value (left) and residuals against predicted value (right)

Model 2 takes logarithms, which turn the multiplication into an addition: log(Total_Sales) = log(Quantity) + log(Price). It fits perfectly (R-squared = 1.000, both coefficients equal to 1). This shows that the regression mostly confirms how the data was recorded. It demonstrates the method correctly, but it is not a discovery about customer behaviour.

7. Business Insights and Recommendations

1. **Concentrate retention on the first year (strong evidence).** 63% of customers with 12 months or less left, against 0% after that. An onboarding programme, early check-ins and first-year offers target the real problem.
2. **Move month-to-month customers to longer contracts (strong evidence).** Their churn rate is 20.6% against 4.3% for one-year contracts. Offer a discount or bonus for switching.

3. **Look into the West region, but gather more data first (suggestive).** West orders are smaller on average (Rs.81,689 against Rs.138,394), yet the evidence is weak after correction. Check whether product mix or pricing explains it before changing strategy.
4. **Watch pricing for customers who leave (suggestive).** Customers who left paid more per month (130 against 112), a small effect (d = 0.35) that did not survive correction. A pricing review for new customers is worth testing, not assuming.
5. **Grow revenue through quantity and price per order.** Order value rises with both, and they are independent. Bundles that raise the number of units per order are a practical lever (about Rs.26,431 per extra unit in this data).
6. **Improve the data.** Record marketing spend so its effect on revenue can be tested, fix TotalCharges so that it agrees with tenure and monthly charges, use one customer ID format in all systems, and collect at least a full year of orders.

8. Limitations

- The sales file has only 100 orders over about three months (April has only 9 days), so small differences are hard to detect and seasonal patterns cannot be studied.
- There is no marketing-spend column, so the marketing-to-revenue question was not tested.
- Order values are not normally distributed. Non-parametric checks agree with the t-tests and ANOVA, but p-values are approximate.
- Ten tests were run. After a Bonferroni correction only the contract and tenure results remain significant.
- The customer and sales files could not be matched customer by customer, so churn results describe the customer file, not the buyers in the sales file.
- TotalCharges is inconsistent with tenure and monthly charges and was not used.
- Correlation and regression show association, not cause and effect.

9. Conclusion

The strongest, most reliable findings are that customers leave mainly in their first year and on month-to-month contracts, and that order value is driven by quantity and price. The differences between products and regions are not statistically clear with this amount of data, and the West-region and monthly-charges results are hints rather than proof. The statistical toolkit (descriptive statistics, distribution checks, correlation, hypothesis tests, confidence intervals and regression) was applied, and its limits were checked rather than hidden.